

## 하이킹 트레일에 대한 편익과 선호도 연구: 선택 실험법을 이용하여

### Benefits and preferences for a hiking trail using a Choice Experiment

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#### Abstract

*The purpose of the study is to identify benefits that hikers seek for and find their preferences for hypothetical hiking-related attributes that reflect the benefits by eliciting hikers' Marginal Willingness to Pay (MWTP). This study employed Choice Experiment (CE), a research method, to convert individuals' utility to MWTPs, which can offer insights about which attributes make hikers better or worse off based on the sizes of MWTPs. 'Degree of Crowding' had the highest MWTP value of all hiking-related attributes, which was estimated to be KRW 5,222 (\$5.14). 'Fitness Equipment,' 'Amenities,' and 'Degree of Plant Variety,' were also significant preferable hiking attributes. Implications and conclusions will be discussed*

**Key Words:** 편익(Benefits), 효용(Utility), 선택실험법(Choice Experiment), 한계지불의사액(Marginal Willingness to Pay)

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## I. Introduction

The hiking population has increased over the last decade in South Korea (hereafter Korea). People's desire for savoring nature with a slower pace has driven a popular walking culture nationwide (Lim & Kim, 2011). As a response to this trend, hundreds of hiking trails have been developed and gained popularity since 2007. Most hiking trails are often called Dulegil which literally means a trail going around the foot of a mountain while not crossing over a hill. It also contains an inherent meaning of 'a natural and comfortable road with a harmony of history, culture, nature, and human' (Lim & Kim, 2011).

Bukhansan Dulegil (BD) comprising a series of hiking trails totaling 71.5-Kilometer was developed in 2010 in Bukhansan National Park (BNP) and is one of the most popular national parks in Seoul, Korea. The original purpose of the BD development is to disperse Bukhansan's heavy volume of hikers to the lower areas of the mountain and reduce overuse of natural resources in BNP. BD has currently received a great deal of attention since it provides hikers pleasurable hiking experiences with a relatively gentle slope and shorter traveling time (Han, July 10, 2014; KNPR, 2013).

Despite of the popularity of trail, a current mass media report indicates that BD's visitation level has noticeably increased by absorbing 1.2 million BNP hikers, which creates hiking-related complaints. The frequently reported issues include a lack of facilities (e.g., public restrooms, rest areas) and unsatisfactory hiking circumstances (e.g., crowdedness) (Cho, Yun, & Im, 2011). According to news articles, accurate demanding forecasting and elaborate development planning had not been conducted and developed in initial stage, which result in the cause of above problems. Besides these, although BD administrators refused to give details of BD financial information, they clarified that they run a budgetary deficit to manage the trails currently, which make it difficult to deal with the stated problems. They concerned that hikers' number will drop significantly if this situation persists.

In order to address the facility and environment issues under the limited

budget circumstance, detailed understanding of hikers' preferences could help BD managers spend their funding conservatively and efficiently. Empirical studies in tourism and leisure contexts have maintained that classification of visitors/tourists contributes to in-depth understanding of their' behavioral preferences or actual choices (Sharpley, 1996; Sarigollu & Huang, 2005; Frochot, 2005; Yannopoulos & Rotenberg, 1999). Especially, classification by utilitarian needs has proved as a powerful procedure to predict tourists' preferences (Formica & Uysal, 1998; Loker & Perdue, 1992; Moscardo, Morrison, Pearce, & O'Leary, 1996). This literature argued that the benefit-based segmentation assists in understanding different segments, which in turn enables tailoring of outdoor recreation opportunities (Anderek & Caldwell, 1991; Locker & Purdue, 1992; Calentone & Jahor, 1984; Lewis & Meadows, 1995; Bruns, Driver, Lee, Anderson, & Brown, 1994; Allen, 1996).

Considering the current shortage of management fund and needs for supplement/reappraisal of hiking facilities and environment in BD, benefit-based segmentation would be helpful to investigate what benefits BD hikers look for and identify how administrators scheme facilities/environment based on need, thereby contributing to maximum efficiency in minimum spending of management fund. To scrutinize the goal, this study will employ Choice Experiment (CE), a research method to elicit economic values of multiple attributes as a form of Marginal Willingness-To-Pay (MWTP). The sizes of estimated MWTPs, referred respondents' utility, become a barometer to measure their preferences over the each attribute. In this study, therefore, the analysis of CE can indicate how BD hikers will be better or worse off when a certain facilities/environment would be given.

## II. Literature Review

### 1. Perceived Benefits

Benefits are defined as “an improvement, the prevention of a worse condition, or a gain” (Tew & Barbieri, 2012, p. 216). Individuals pursue different benefits from their

product use or activities (Allen, 1996; Woodside & Jacobs, 1985). Understanding perceived benefits often provide knowledge about market segments and associated information such as market sizes, consumer preferences, and expenditure patterns (Palacio, 1997). Traditionally, markets were divided by socio-demographic, geographic, behavioral, and psychological factors; however, knowledge about consumers' benefits pursued have been considered a powerful indicators in predicting their behavioral patterns (Haley, 1985; Young, Ott & Feigin, 1978).

In the context of tourism and outdoor recreation, researchers also have focused on perceived benefits to better understand tourists/recreationists (Palacio, 1997). Numerous studies have categorized tourists/recreationists by their benefits sought and their heterogeneous behavioral characteristics. For example, Loker and Perdue (1992) examined whether it is feasible to segment a nonresident tourist market based on pursued benefits from vacations. They found six distinct market segments which includes naturalists, nondifferentiators, family/ friend-oriented, excitement/escape, pure excitement seekers, and escapists. Frochot (2005) also divided rural tourists in two Scottish locations into four clusters based on the benefits they sought. The results indicated that each segment was distinguished from one another in terms of the activities engaged. Palacio (1997) applied this approach in the ecotourism context. He identified four segments of ecotourists among visitors to Belize using perceived benefits to segment ecotourists instead of defining them by participation in activities (Shin et al., 2001; Loker & Perdue, 1992; Frochot, 2005; Palacio, 1997; Ho and his colleagues, 2005; Hunt & Ditton, 2010).

## 2. Stated Preference: Choice Experiment and Its Use in a Recreation/Tourism Context

The stated preference (SP) approach is a systematic method to elicit individuals' preferences for particular products/services or estimate the economic value of non-market goods. Basic logic of the SP is that researchers provide respondents with hypothetical scenarios describing potentially real changes in research subjects/circumstances and ask what individuals say in the situations rather than what they actually do in real situations. Therefore, the main advantage of SP is to manipulate qualities or quantities of research attributes, which eventually enables researchers to estimate economic values for non-market

goods or preferences between the manipulations (Han & Lee, 2008; Kim et al., 2007; Lee & Han, 2002; Mmopelwa, Kgathi, & Molefhe, 2007; Reynisdottir et al., 2008; Han & Lee, 2008; Lee & Han, 2002; Mmopelwa, Kgathi, & Molefhe, 2007).

Discrete Choice Experiment (DCE) is a representative SP technique, specializing in finding which attributes are important in determining individuals' preference for products, services or programs, each of which is characterized by multiple attributes. Through DCE, researchers can estimate the benefit which an individual receives from consuming one more unit (level) of a goods as a form of individuals' MWTP. Based on the sizes of MWTPs estimated, researchers can find ranking of attributes preferred, thereby ultimately offering insights about which attributes make respondents better or worse off based on changes of attributes given (Crouch & Louviere, 2001).

DCE has been widely used in the context of tourism and outdoor recreation research since the beginning of 21st century. Researchers consider DCE as highly useful in that it can capture "systematic heterogeneities of different groups of tourists'/recreationists' preferences for a variety of multi-attribute tour/outdoor products and services" (Hanley, Wright, & Adamowicz, 1998; Correia, Santos, & Barros, 2007; Oh, Ditton, & Riechers, 2007; cited in Lee, 2013, p. 21). For example, Oh and Ditton (2006) identified how each segment of anglers perceived trade-offs among regulatory attributes and found heterogeneous preference levels (e.g., entrance fee, number or size of fish allowed to catch). Also, Riganti and Nijkamp (2008) employed DCE to develop an appropriate management policy at a mature cultural tourism destination by examining tourists' heterogeneous preferences. Considering the advantage of CE, the main purposes of the study is to investigate preferences for facilities and environments by estimating and comparing their MWTPs.

### III. Method

#### 1. Pre-test to Identify BD Hikers' Benefits Sought

This study carried out pre-test with BD hikers to uncover recreational

benefits that they pursue. The pre-test helped to recognize what benefits BD hikers pursue, which eventually assisted in determining what facilities/environments reflect the benefit before the application of the DCE analysis. Based on the three benefit studies in recreational and outdoor settings (i.e., McColl & Reilly, 1993; Driver & Brown, 1987; Palacio, 2010), a benefit measurement scale used in the study was modified to measure BD hikers' benefits. Two hundred survey questionnaires were distributed and 189 were collected. Forty five of them were eliminated due to severe data missing.

Following the data collection, exploratory factor analysis (EFA) using principal component analysis with varimax rotation was performed on the 15 benefit items. Table 1 illustrates the names of each dimension, numbers of items, mean, standard deviation, factor loadings, and communality. The Kaiser-Meyer-Olkin measure (KMO) and Bartlett's Test of Sphericity were used to examine the appropriateness of factor analysis as shown in Table 1.

〈Table 1〉 Eigenvalues, Variance explained, and Cronbach's Alphas of Benefit items

| Kaiser-Meyer-Olkin       |            | Measure of Sampling adequacy |       | 0.806      |              |          |  |
|--------------------------|------------|------------------------------|-------|------------|--------------|----------|--|
| Bartlett's               |            | Test of Shpericity           |       | 4608.879   |              |          |  |
| Factor                   | # of Items | Mean                         | SD    | Eigenvalue | Variance (%) | $\alpha$ |  |
| Learning Nature          | 4          | 3.164                        | 1.131 | 3.723      | 23.267       | .922     |  |
| Escaping from daily life | 5          | 3.301                        | 0.771 | 3.098      | 19.360       | .944     |  |
| Health Improvement       | 3          | 3.175                        | 0.936 | 2.659      | 16.621       | .903     |  |
| Socializing with others  | 3          | 2.735                        | 1.235 | 2.640      | 16.498       | .855     |  |

† Note1: N = 155

Note2: Extraction method: Principal Component Analysis

Note3: Rotation Method: Varimax with Kaiser Normalization

Four factors were found and labeled as follows: Factor 1= Learning Nature (4 items; alpha=.922); Factor 2 = Escaping from daily life (5 items; alpha=.944); Factor 3= Health Improvement (3 items; alpha=.903); Factor 4 = Socializing with others (3 items; alpha=.855). According to the factor' means, respondents regarded Factor 2 (Escaping from daily life) as being the most important benefit, Factor 4 (Socializing with others) as the least important benefit.

## 2. List of Attributes and Levels

To investigate the systematic preferences for BD facilities/environment, DCE was primarily used in this study. It requires respondents to choose one among two or more competing alternatives that vary along several attributes' characteristics. Therefore, attributes and their corresponding characteristics' levels should be identified prior to DCE data collection and analysis (Louviere & Timmermans, 1990, Louviere et al., 2000; Birol, Karousakis, & Koundouri, 2006). Consulting with experts in a certain field is one of typical approaches to identify and define significant attributes. Once the attributes were defined, corresponding levels of the attributes were also determined through the same discussion process. In consideration of factors found from pre-test analysis results, 4 attributes which could reflect the benefits were determined through the discussion with BD administrators. In addition to the four attributes, this study includes donation as the monetary attribute, which contributes to deriving MWTPs. After 5 attributes were determined, three rank levels were assigned to the attributes by the same discussion process. This plays an important role in identifying heterogeneous preferences for hiking-related attributes because theory behind CE assumes that respondents will compare the levels of attributes and choose a profile including preferred levels of attributes. Table 2 shows the attributes and corresponding levels.

〈Table 2〉 Five hiking-related attributes and corresponding levels

| Attributes                 |           | Levels                                                    |
|----------------------------|-----------|-----------------------------------------------------------|
| Degree of Crowding         |           | Status quo (=a bit crowded)                               |
|                            |           | 1.5 times less crowded than status quo                    |
|                            |           | 2 times less crowded than status quo                      |
| Fitness Equipment          |           | Stay with the current equipment number in a route         |
|                            |           | 1.5 times increase of current equipment number in a route |
|                            |           | 2 times increase of current equipment number in a route   |
| # of Rest area & cafeteria | Amenities | Stay with the current amenity number in BD                |
|                            |           | 1.5 times increase of current amenity number in BD        |
|                            |           | 2 times increase of current amenity number in BD          |
| Degree of Plants Variety   |           | Stay with the current plant species number in BD          |
|                            |           | 1.5 times increase of current plant species number in BD  |
|                            |           | 2 times increase of current plant species number in BD    |
| Donation                   |           | 5,000 KRW                                                 |
|                            |           | 10,000 KRW                                                |
|                            |           | 15,000 KRW                                                |

† Note: US \$1 is equivalent to 1,016 Korean Won

### 3. Choice Sets and Experimental Design

A profile is a set of an attributes' level. In this study, two profiles and one opt-out profile ("I like maintaining the status quo") consist of a choice set (See Table 3). When a respondent may not like either A or B profile, he or she may select an opt-out profile. Eight choice sets were provided with respondents and they were asked to choose only one profile that they prefer in each choice.

Since there are 5 attributes and 5 corresponding levels each, 243 ( $3 \times 3 \times 3 \times 3 \times 3$ ) potentially possible profiles (called full factorial design) that could be derived; however, it is unrealistic to inquire respondents' favorites for all 243 cases in a survey questionnaire. Instead of providing entire 243 combinations, there is a methodological technique, called a fractional-factorial design, for extracting a smaller number of representative of profiles but, at the same time, drawing the most efficient linear design. In other words, this technique enables researchers to estimate attributes' main effects with carefully chosen subsets of the experimental runs of a full factorial design. The Statistical Analysis System (SAS) was used for the fractional-factorial design and total 8 choice sets were derived. Table 3 shows an example, one of the eight fractional factorial choice sets:

〈Table 3〉 An example of a choice set sent to respondents

| Attributes                         |  | Profile A                                                |  | Profile B                                              |  | I would not                   | take A |
|------------------------------------|--|----------------------------------------------------------|--|--------------------------------------------------------|--|-------------------------------|--------|
|                                    |  |                                                          |  |                                                        |  | or B                          |        |
| Degree of Crowding                 |  | Status quo (a bit crowded)                               |  | 50% less crowded than what it has now                  |  | I like keeping the status quo |        |
| Fitness Equipment                  |  | 2 times increase of current equipment number in a route  |  | Stay with the current equipment number in a route      |  |                               |        |
| Amenities (Rest area or cafeteria) |  | 2 times increase of current amenity number in BD         |  | 1.5 times increase of current amenity number in BD     |  |                               |        |
| Degree of plants variety           |  | 1.5 times increase of current plant species number in BD |  | 2 times increase of current plant species number in BD |  |                               |        |
| Donation                           |  | 15,000 KRW                                               |  | 10,000 KRW                                             |  |                               |        |

#### 4. Theories of Choice Experiment

Two main economic consumer theories behind DCE build the foundation of the DCE: Lancaster's Characteristics Approach (LCA) and Random Utility Maximization Theory (RUT) (Adamowicz, Louviere, & Swait, 1998). First, LCA explains that individuals derive utility not only from a bundle of goods or service but also the characteristics of these goods (Gravells & Rees, 1992). RUT, on the other hand, accounts for individuals' rational decision process using the concept of utility components (Manski, 1977). According to the RUT theory, individuals are more likely to choose a good that maximizes utility among the given alternatives. He partitioned the utility function into observable and non-observable components. The following equation expresses the utility function for an alternative  $j$  of the individuals:

$$U_{ij} = V_{ij} + \varepsilon_{ij} \quad (1)$$

Where  $U$  stands for  $i$  individuals' indirect utility when choosing  $j$  alternatives.  $U$  consists of  $V$ , observable utility from goods'/services' characteristics and, unobservable utility also called stochastic component. The unobservable utility represents individuals' idiosyncrasies of choice tastes, which elucidating unexplained variations of individuals' choice actions (Apostolakis & Jaffry, 2005).

According to the theoretical logic of RUT,  $i$  individuals select  $j$  alternative when the utility obtained from selecting  $j$  good is greater than utility for others,  $k$ , probability of individual  $i$  choosing alternative  $j$  as follows:

$$\begin{aligned} P_{ij} &= \Pr \{ U_{ij} \geq U_{ik} \} \\ &= \{ V_{ij} - V_{ik} \geq \varepsilon_{ij} - \varepsilon_{ik}; \quad j \neq k, j, k \in J_i \} \end{aligned} \quad (2)$$

If the random components ( $\varepsilon_{ij}$  and  $\varepsilon_{ik}$ ) are assumed to be Gumbel-distributed and independently and identically distributed across alternatives, the probability can result in the conditional (or multinomial) logit model as following Eq (3):

$$P_{ij} = \exp(\mu V_{ij}) / \sum \exp(\mu V_{ik}) \quad (3)$$

where  $\mu$  is a scale parameter, which is typically set to one for parameter estimation. For the parameter estimates, log maximum likelihood is applied. Following Eq (4) shows log likelihood function:

$$\ln L = \sum_{i=1}^n \sum_{j=1}^m \delta_{ij} \ln P_{ij} \quad (4)$$

where  $\delta_{ij}$  is a dummy variable such that  $\delta_{ij} = 1$  if alternative  $j$  is chosen and  $\delta_{ij} = 0$  otherwise. Once model parameters are estimated, marginal willingness to pay, a main purpose of Choice Experiment, can be derived as follows:

$$\text{Marginal Willingness to pay} = \frac{\beta_{\text{non-cost-attribute}}}{-\beta_{\text{cost-attribute}}} \quad (5)$$

## 5. Survey and Data Collection

On-site survey was carried out using a systematic sampling method (every 5th participant who was over the age of 18) at end-point of Routes of 'Pyeongchang' and 'Sullye-gil', the most visited routes, in BD on December in 2013. Only one representative family member and respondents who experienced or completed BD hiking were allowed to participate in the survey. Researchers and well-trained graduate students described the study background, purpose, and guidelines to answer CE questions to minimize misinterpretation of the choice scenarios. Also, questions about socio-demographics were asked. Five hundred twenty three respondents were willing to join the survey and 400 questionnaires were used after eliminating 99 incomplete responses and 24 invalid responses. As a result, a total of 3,200 (400 respondents \*8 choice sets) choice responses were used for the data analysis.

## IV. Results

### 1. Estimation Results of the Study Model

Table 4 shows parameter estimates for five main attributes: ‘Degree of Crowding,’ ‘Fitness Equipment,’ ‘Amenities,’ ‘Degree of Plant Variety,’ and ‘Donation.’ Hypothesis tests were conducted at .05% level through log maximum likelihood and all main attributes were statistically significant. Also, the alternative specific constant (ASC), the utility that individuals obtain when choosing either A or B alternatives over the opt-out choice, was included in the study model. ASC was determined as positively significant variables, indicating that respondents will be better off when choosing either A or B profile over the opt-out choice. Table 4 shows the estimation results of the main variables.

〈Table 4〉 Estimation results of the main attributes

|                          | Estimates   | Standard Error | Z      | Pr>   |
|--------------------------|-------------|----------------|--------|-------|
| Degree of Crowding       | .5066**     | .0360          | 14.07  | 0.000 |
| Fitness Equipment        | .2479**     | .0325          | 7.63   | 0.000 |
| Amenities                | .3923**     | .0410          | 9.56   | 0.000 |
| Degree of plants variety | .4042**     | .0332          | 12.16  | 0.000 |
| Donation                 | -.0000833** | 6.67E-06       | -12.50 | 0.000 |
| ASC                      | .4187**     | .1998          | 2.10   | 0.036 |
| Log-Likelihood           | -4855.4467  |                |        |       |

† Note 1: \*\* indicates significance at .05 level

Note 2: ASC stands for Alternative Specific Constant

According to the table 4, the estimated coefficient of “Degree of Crowding” is .5066 with a corresponding p-value of <.001, representing the log-odds when a level of the attribute moves to uncrowded status. Other variables, with the exception of ‘Donation,’ are all positive and significant, implying that respondents

were likely to construct additional fitness equipment and amenities, and increase the number of plants that BD hikers can encounter during hiking. The parameter of donation, on the other hand, illustrates that an increase of donation price was not preferred by respondents.

Also, this study examines whether those who seek different benefits hold heterogeneous preferences regarding the four attributes. To investigate this, the model including four interaction terms (composed by levels of attributes and factor means) was examined. Table 5 shows the estimation results of the MNL model with interaction terms

〈Table 5〉 Estimation results of the MNL model with interaction terms

|                          | Estimates   | Standard Error | Z      | Pr>   |
|--------------------------|-------------|----------------|--------|-------|
| Degree of Crowding       | .4360**     | .0498          | 8.72   | 0.000 |
| Fitness Equipment        | .1600**     | .0505          | 3.17   | 0.002 |
| Amenities                | .3307**     | .0511          | 6.46   | 0.000 |
| Degree of Plants Variety | .3027**     | .0618          | 4.89   | 0.000 |
| Donation                 | -.0000835** | 6.67e-06       | -12.50 | 0.000 |
| ASC                      | .4151**     | .1999          | 2.08   | 0.038 |
| DC_int                   | .0226**     | .0109          | 2.05   | 0.041 |
| FE_int                   | .0277**     | .0122          | 2.27   | 0.023 |
| Amenities_int            | .0227**     | .0112          | 2.02   | 0.043 |
| DPV_int                  | .0309*      | .0158          | 1.95   | 0.052 |
| Male                     | -.0140      | .0507          | -.28   | 0.782 |
| Edu                      | .0051       | .0253          | .20    | 0.839 |
| Income                   | .0007       | .0164          | .04    | 0.966 |
| Age                      | .0005       | .0021          | .24    | 0.809 |
| Log-Likelihood           | -4849.5807  |                |        |       |

\*\* indicates significance at .05 level

\* indicates significance at .1 level

Note 1: DC\_int: Interaction term between “Degree of Crowding” and mean of Factor 2; FE\_int: Interaction term between “Fitness Equipment” and Factor 3; Amenities\_int: Interaction term between “Amenities” and mean of Factor 4; DPV\_int: Interaction term between “Degree of Plants variety” and Factor 1

According to the above results, those of who seek ‘Escaping from daily life’ benefit are more likely to be isolated from others, who seek ‘Health’ benefit are more likely to have fitness equipment, who seek ‘socializing with others’ benefit are more likely to have more rest areas or cafeteria, and who seek ‘learn’ benefit

are more likely to experience a variety of species of plants. Marginal Willingness-To-Pay (MWTP) can be calculated under the logic of Eq (5). Table 6 indicates MWTPs of attributes.

〈Table 6〉 Marginal Willingness-To-Pay

|                          |                                    |
|--------------------------|------------------------------------|
| Degree of Crowding       | 5,222 KRW [4,750 KRW - 5,684 KRW ] |
| Fitness Equipment        | 1,916 KRW [1,652 KRW - 2,256 KRW]  |
| Amenities                | 3,960 KRW [3,542 KRW - 4,339 KRW]  |
| Degree of Plants Variety | 3,625 KRW [3,115 KRW - 3,852 KRW]  |
| ASC (Baseline tour)      | 4,971 KRW [4,623 KRW - 5,260 KRW ] |

Note: The numbers in brackets are 99% confidence intervals, obtained using the methods of Krinsky and Robb (1986) and based on 1,000 random draws.

As shown in Table 6 above, in terms of the MWTP sizes, respondents are likely to pay 5,222 KRW (\$5.14), 1,916 KRW (\$1.89), 3,960 KRW (\$3.90), and 3,625 KRW (\$3.57) for one more level increase of attributes respectively. In the same way, BD hikers are willing to pay 4,971 KRW (\$4.89) for going from on a ‘status quo to’ ‘supplement of facilities or environment improvement.’

## V. Discussion and Conclusions

Current management fund deficit and needs for improvement hiking facilities and improvement of physical environment in BD provide the justification of the research on understanding the respondent’s preferences over the hiking-related attributes. For more scientific approach to the study issue, the study identified **benefits** that BD hikers seek for and predetermined hiking-related attributes (facilities/environment conditions) reflecting the benefits. Once the attributes and levels were defined, CE can draw parameter estimates of the chosen facilities/environment conditions and ultimately can assess respondents’ sizes of MWTPs. **Considering the definition of ‘marginal utility,’ the results can become an indicator about how much BD hikers’ welfare will be better off when consuming one more unit of attributes.**

This study contributes to limited literature on estimation of economic values

of hiking facilities/environment using CE. Literatures have actively focused on the relationship between hikers' psychometric factors and behaviors; however, few studies have investigated economic values of multi-attributes in leisure settings by applying CE method. Considering the current shortage of management fund and need for systematic improvements of hiking facilities/environments, the study guides how administrator schemes facilities/environments make hikers being better off in the given conditions.

Coefficients of all four attributes are statistically significant, indicating that given four attributes (except the donation) play an important role in increasing chances of visiting BD. Among the 4 attributes given, attribute of 'Degree of Crowding' shows the highest MWTP, representing that BD hikers place the most priority on improvement of current visitation level. This result can reflect the unfavorable visitation condition. Given the news articles and official data, the annual number of BD hikers in 2013 has reached 10 million and even it is absorbing a large numbers of BNP hikers, which seems like excess of BD capacity. Considering the MWTP size of 'Degree of Crowding,' and current heavy visitation level, it is worthwhile for BD managers to consider restriction policy regarding visitation level. According to the carrying capacity conceptual framework in a leisure context, numbers of encounters during the hiking is inversely correlated with their satisfaction (Graefe, Vaske, & Kuss, 1984; Robson, & Eagles, 2002). For instance, Butler (1996) argued that people negatively perceive the destination which fails to manage their carrying capacity. Davis (1963), Fisher and Krutilla (1972) and Newmann, Manning, Dennis, and McKonly (2005) also showed that adequately managed carrying capacity conditions can enhance the economic value of leisure facilities. Regarding the findings of these previous researches, this research externally validated the results which carrying capacity is regarded as essential attribute to enhance the value of leisure resource. In this respect, BD managers need to monitor hiking congestion level continuously and to make an effort to control the visitation levels and provide pleasant hiking condition.

Also, respondents would like to set up or experience more places for 'Socializing with Others,' 'Fitness Equipment,' and 'Degree Plants Variety' in the BD. The CE results show that these three attributes also have relatively high

MWTPs. As Robson and Eagles (2002) argued that shapes of facilities and varieties of hiking opportunity should be considered depending on types of leisure settings and hikers, BD administrators, therefore, fully understand what to; where to; how to build them in BD.

The study is not free from the limitation. Theoretically, DCE hypothesizes the Independence of the Irrelevant-Alternatives property (IIA), implying that individuals' choice probability across one alternative does not depend on the choice probability of any other alternative (Ben-Akiva & Lerman, 1985). It is a known fact that this assumption has been criticized because the DCE is unable to deal with the unobserved heterogeneity of the individuals' preferences. Therefore, calibration approaches have been suggested to mitigate the IIA assumption.

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2015년 07월 17일 수정본 접수

2015년 07월 20일 최종 수정본 접수 및 게재확정

3인 익명심사 필